EXPERIMENT-5

TITLE: DCL (Data Control Language)

Objective: The primary objective is to enforce database security and manage user access by controlling permissions to database objects.

Aim:

To implement commands like GRANT and REVOKE.

Theory:

DCL deals with rights, permissions, and other controls of the database system.

SQL Code:

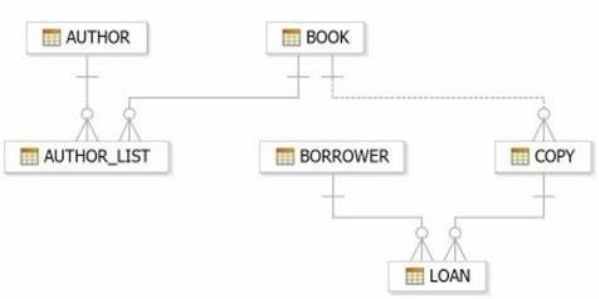
```
GRANT SELECT, INSERT ON employee TO 'user1'@'localhost';
REVOKE INSERT ON employee FROM 'user1'@'localhost';
```

Output:

The expected output confirms successful execution of SQL commands and matches the theoretical concept.

Result:

Successfully executed as per the aim of the experiment.



Exercise

1. Create table AUTHOR = {Author_ID , Lastname, Firstname, Email, City, Country}

Where:

Author_ID – text data type, 5 characters, primary key

Lastname – text data type, 15 characters, not null

Firstname – text data type, 15 characters, not null

Email – text data type, 40 characters,

City – text data type, 15 characters,

Country – text data type, 15 characters,

2. Create Table BOOK={ Book_ID, Book_Title, Copies)

Where:

Book_ID – text data type, 5 characters Primary Key Start with Character **B**

Book_Title - Text data Type Not Null

Copies- No.of copies Data Type int always greater the 2

3. Create table AUTHOR_LIST = {Author_ID , Book_ID , Role}

Where:

Author_ID – text data type, 5 characters, referenced by Author_ID from AUTHOR table

Book_ID – text data type, 5 characters

Role – text data type, 15 characters

and primary key is Author_ID, Book_ID

4. Add four records in each tables AUTHOR, BOOK, BOOK_LIST.

5. Alter structure of table AUTHOR_LIST add the field Publisher data type of 30 Character.

6. Create two new database users named manager_user and staff_user.

7. Grant SELECT, INSERT, and UPDATE privileges on the AUTHOR table to manager_user.

8. Grant only SELECT privileges on the BOOK table to staff_user.

9. Revoke the INSERT privilege from manager_user and verify the restriction.